Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-9550
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Script-Controlled ACL solution can be extended to support larger numbers of users and records, stronger access-control requirements, and additional features in the future. It captures the roadmap for enhancing the ServiceNow ACL implementation beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Access rules currently depend on a small set of custom roles and branch-based conditions.	May require additional role and condition management as users and departments increase.	High
2	The current implementation focuses on record-level READ, CREATE, WRITE and DELETE ACLs.	More complex field-level or application-level security may be needed for larger systems.	Medium
3	Testing is performed with selected users and impersonation.	Large-scale permission testing may become time-consuming as roles and users grow.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Custom roles bb1–bb4 are used for controlled access.	Introduce a structured role hierarchy and reusable access rules for larger user groups.
2	Data Storage	Institution Details table stores student and related details.	Expand table design and indexing as record volume grows while maintaining secure access.
3	Performance	ACL scripts evaluate roles and field values during access.	Optimize conditions and scripts, reduce unnecessary checks, and test ACL performance at scale.
4	Security	READ, CREATE, WRITE and DELETE ACLs control record access.	Add more granular field-level controls, auditing and periodic permission reviews.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Refine existing READ, CREATE, WRITE and DELETE ACL rules and test role combinations.	Short Term	More consistent and maintainable access control.
Phase 2	Add field-level security and expanded branch/department-based access rules.	Medium Term	More granular protection of sensitive fields.
Phase 3	Introduce stronger auditing, access reviews and automated permission testing.	Medium Term	Improved monitoring and easier security validation.
Phase 4	Scale the solution for additional tables, departments, users and future ServiceNow applications.	Long Term	Reusable and scalable record-level security framework.